LAB5 Feature Tracking Comparison

VNAV2021 code | Ubuntu 20.04 / ROS Noetic | OpenCV 3.4.2

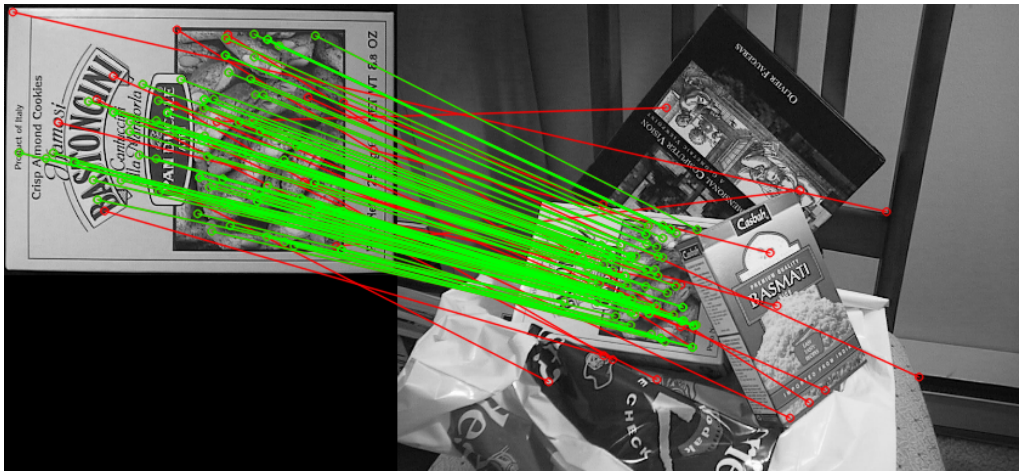
Methods and scoring

SIFT and SURF use FLANN two-neighbor matching and the 0.8 distance ratio. ORB and FAST+BRIEF use Hamming-distance brute force with the same ratio. Harris+LK tracks corners by pyramidal optical flow. All five methods use fundamental-matrix RANSAC with a 3-pixel threshold and 0.99 confidence to count geometric inliers.

Bundled image pair

box.png to box_in_scene.png. SIFT descriptors have 128 float values; SURF has 64 float values; ORB and BRIEF have 32 bytes.

Method	Keypoints 1	Keypoints 2	Raw matches	Good matches	Inliers	Inlier ratio
SIFT	603	969	603	99	79	79.8%
SURF	997	1822	997	129	85	65.9%
ORB	500	500	500	13	7	53.8%
FAST+BRIEF	382	462	382	8	7	87.5%

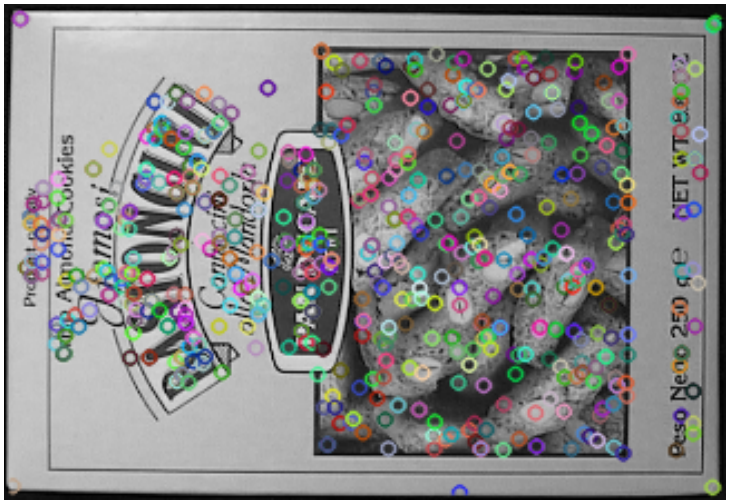


Observation. The large viewpoint and scale change makes this pair harder than adjacent video frames. SURF gave the most RANSAC inliers (85), while SIFT had the higher inlier fraction (79.8%). At the supplied 500-feature, one-level setting, ORB retained only 13 unambiguous matches and 7 inliers. FAST+BRIEF had 8 ratio-filtered matches and 7 inliers, so its high inlier fraction has a small denominator. Green lines in the figure are inliers; red lines are rejected matches.

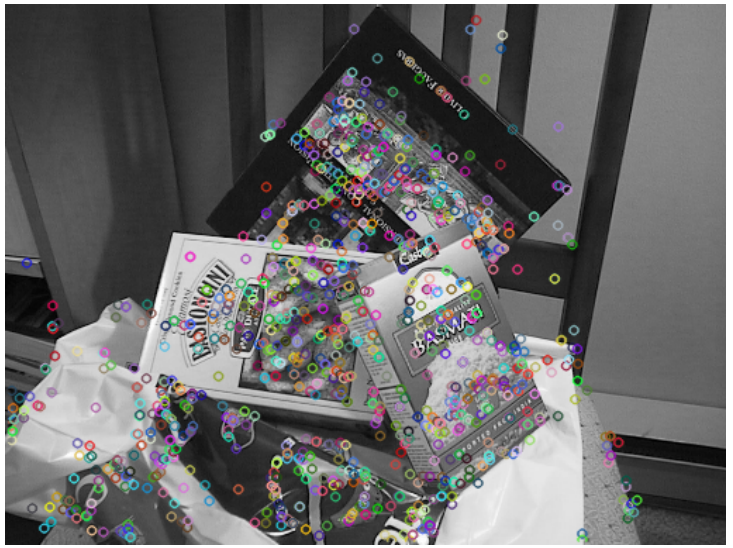
SIFT detection and matching figures

The two-image run saves these figures directly from the tracker. There are 603 and 969 keypoints, 603 raw nearest-neighbor matches, and 99 matches after the ratio test.

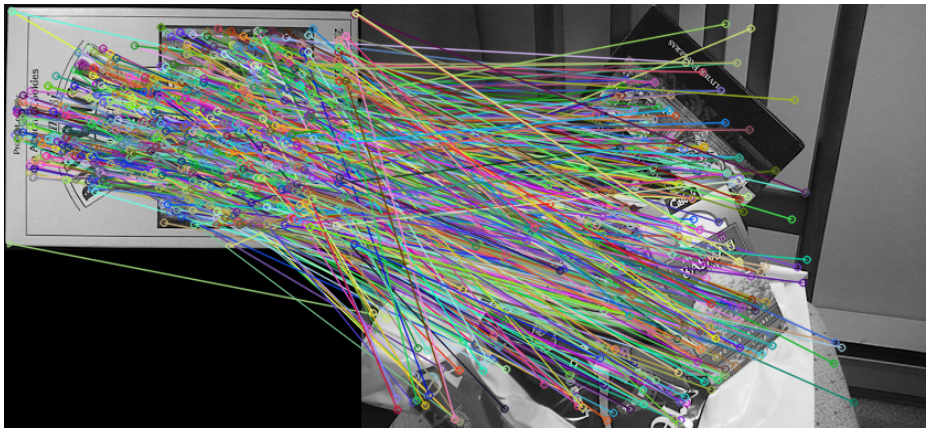
1. SIFT keypoints in box.png



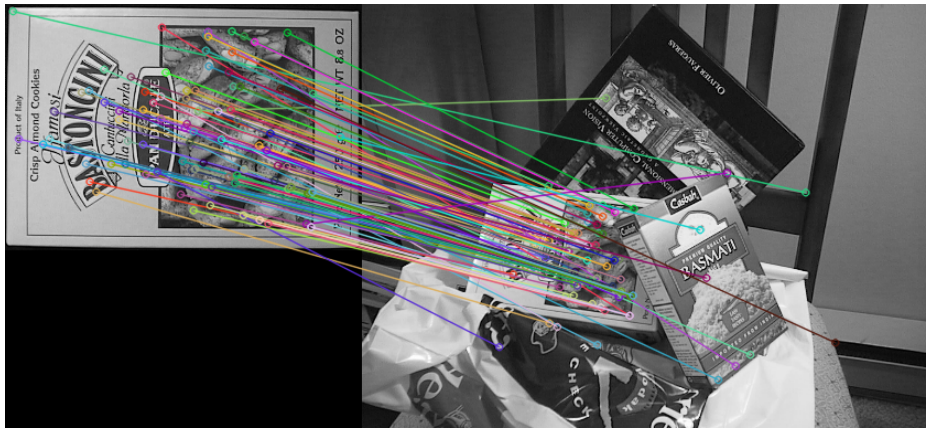
2. SIFT keypoints in box_in_scene.png



3. Raw nearest-neighbor matches



4. Matches after the 0.8 ratio test



The ratio test removes ambiguous correspondences before geometric verification. The colored inlier/outlier plot appears on page 1.

Real flight sequences

Each table averages all adjacent RGB-frame pairs. Avg. ratio is the mean per-pair inlier fraction. Playback was slowed as needed so the VM did not drop images.

Dataset 1: 30fps room sequence

30fps_424x240_2018-10-01-18-35-06.bag | 1584 RGB frames | 0.6x playback

Method	Avg. KP1	Avg. KP2	Avg. matches	Avg. good	Avg. inliers	Avg. ratio	Pairs
SIFT	403.5	403.6	403.5	258.4	249.0	95.7%	1583
SURF	466.9	467.0	466.9	334.2	319.9	95.3%	1583
ORB	315.7	315.7	315.7	230.2	226.0	97.9%	1583
FAST+BRIEF	274.2	274.2	274.2	227.9	221.0	96.7%	1583
Harris+LK	102.2	102.2	100.9	100.9	100.8	99.9%	1583

Dataset 2: smooth trajectory

vnav-lab5-smooth-trajectory.bag | 1386 RGB frames | 0.35x SIFT/SURF; 0.6x ORB/FAST/LK

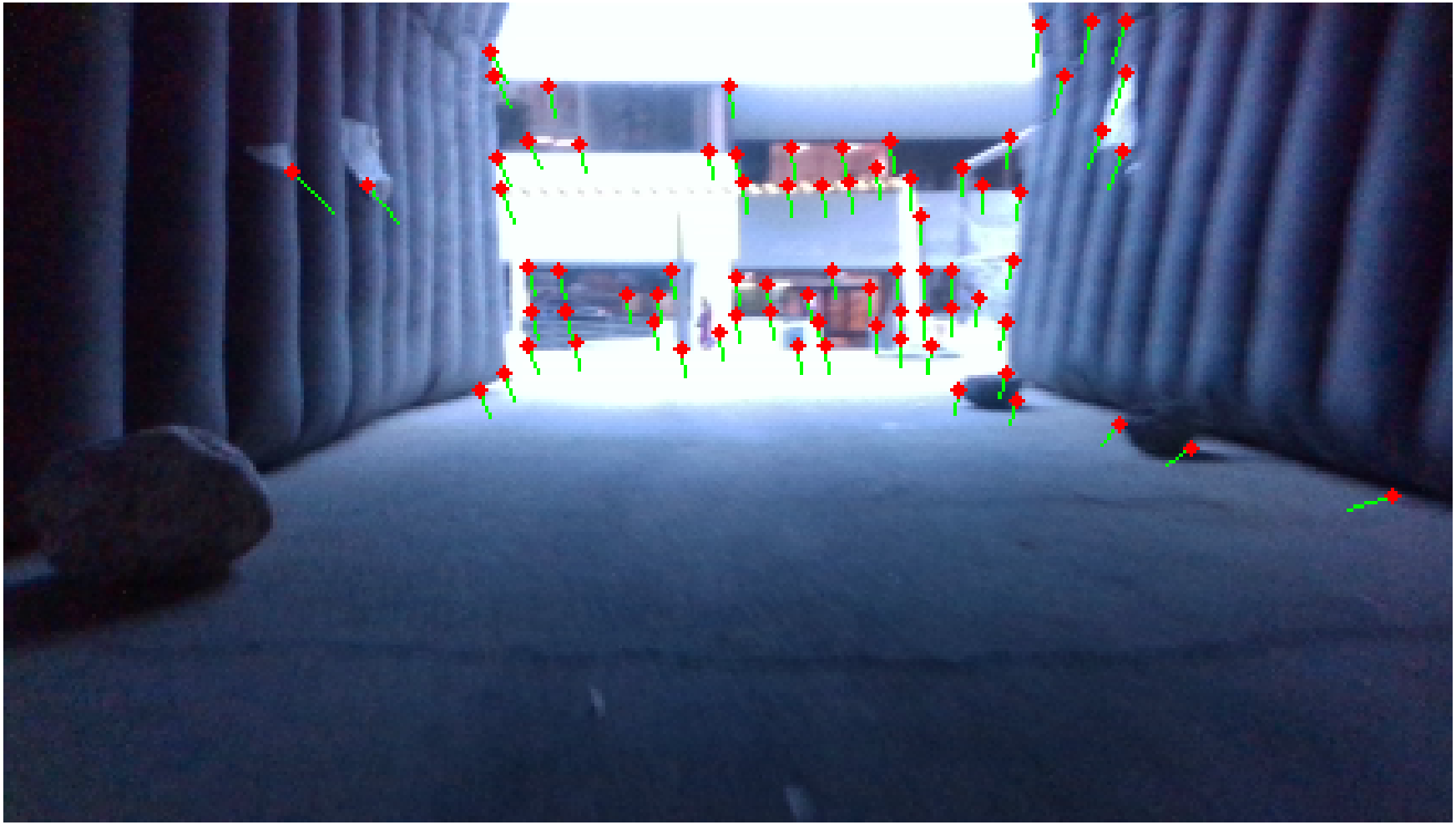
Method	Avg. KP1	Avg. KP2	Avg. matches	Avg. good	Avg. inliers	Avg. ratio	Pairs
SIFT	147.8	147.9	147.8	92.9	80.7	86.1%	1385
SURF	1191.8	1192.9	1191.8	714.1	609.6	85.5%	1385
ORB	278.8	279.1	278.8	166.8	151.2	90.8%	1385
FAST+BRIEF	268.3	268.6	268.3	188.7	174.7	92.3%	1385
Harris+LK	70.6	70.7	68.9	68.9	64.2	93.9%	1385

Interpretation and limits

SURF led average inliers on both sequences (320 and 610). FAST+BRIEF reached 221 and 175, close to ORB's 226 and 151. Harris+LK had the highest inlier fraction but fewer tracks; it assumes small motion and roughly constant brightness. Runtime was not measured, so no speed ranking is claimed.

Harris + LK track example

Real RGB frames 300 and 305 from Dataset 1. The five-frame gap makes short tracks visible; 75 of 78 previous-frame corners were tracked in this illustration.



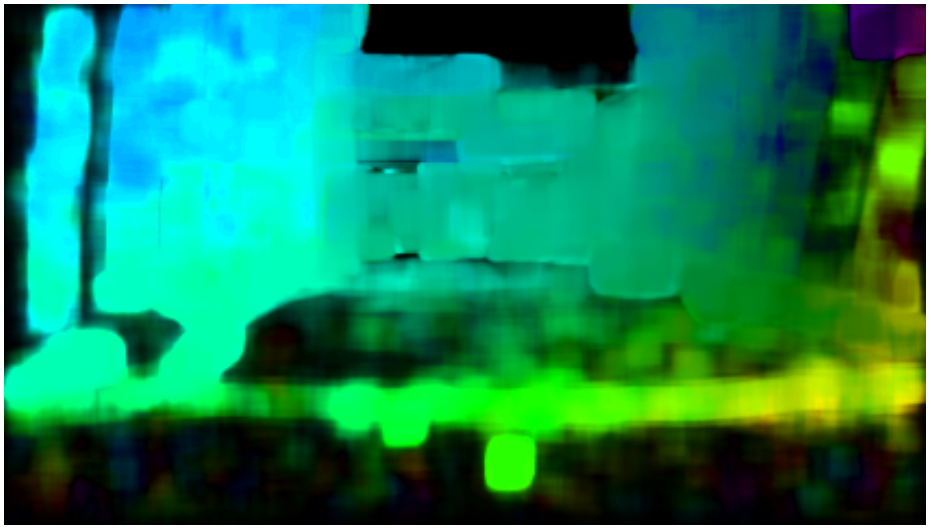
Green segments show motion between the two frames; red dots mark tracked positions in the later image. The full-dataset LK statistics on page 3 use consecutive frames, as do the other methods.

Assignment and data: MIT 16.485 VNAV LAB5 Exercises (2023), vnav.mit.edu/labs_2023/lab5/exercises.html; github.com/MIT-SPARK/VNAV-lab-data.

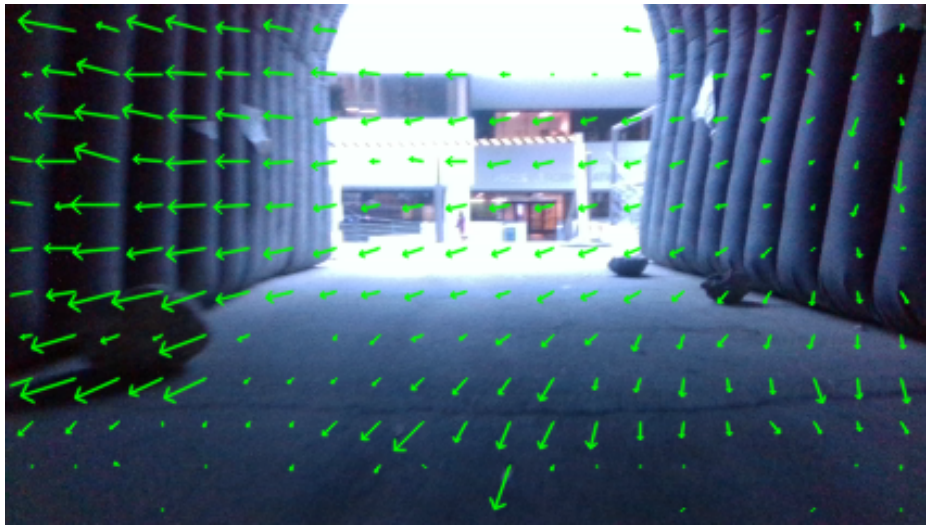
Optional Deliverable 8: dense optical flow

Farneback flow on consecutive RGB frames from Dataset 1. 1583 adjacent pairs were processed; the selected pair is #300.

Direction and magnitude (HSV)



Sampled flow arrows on the RGB frame



Interpretation

Each pixel receives a 2D displacement estimate. Hue shows motion direction; brightness increases with magnitude and saturates at 3.0 pixels/frame. The arrows sample the same field every 20 pixels and are drawn four times longer for legibility. The selected pair has mean motion 1.93 pixels and maximum 7.73 pixels. Across the processed sequence, the average per-pair mean motion is 2.48 pixels/frame. Flow is visible across surfaces, whereas Harris+LK samples only corner tracks. Bright or low-texture regions and occlusions can make local flow less reliable.